

Jahn–Teller Effect in Isomerization of Cubane

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Cubane isomerizes to cuneane via breaking two C–C bonds and forming other bonds in the presence of Ag(I) Lewis acid catalysts [1]. A substituted cubane yields product isomers, and their formation ratio depends on the introduced substituents. The reason for such a substituent effect has been veiled. In the present study, we determine a reaction path of the isomerization using DFT calculations and analyze the obtained reaction mode. In the early stage of the reaction, the charge transfer from cubane to Ag(I) causes the Jahn–Teller effect, and further, the presence of the low-lying excited state gives rise to the pseudo-Jahn–Teller effect. These effects produce one of the stable structures by breaking a C–C bond. Cubane can be thermally transformed between such stable structures while passing through the structures with two C–C bonds in a skew position simultaneously cleaved, assisting the conversion to cuneane. This result implies that introducing substituents warps the potential energy surface asymmetrically and stabilizes the structure with certain two C–C bonds broken, consequently changing the formation ratio of the product isomers.

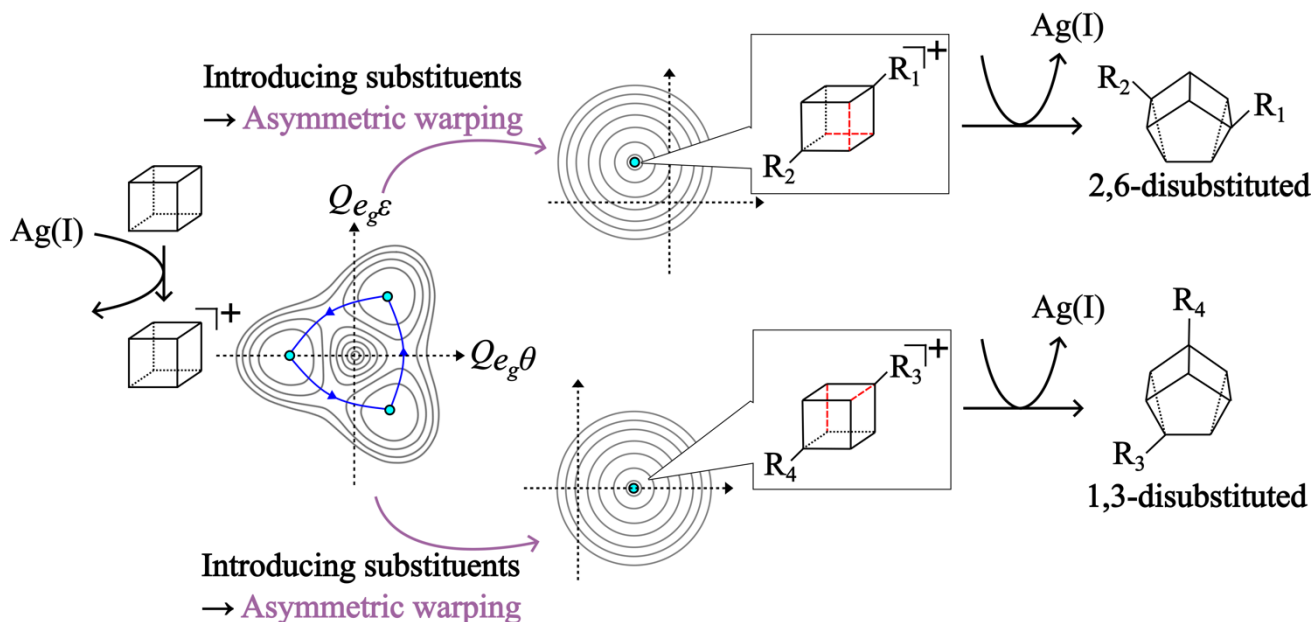


Fig. 1. The schematic diagram of isomerization reaction paths.

Reference

- [1] H. Takebe and S. Matsubara, *Eur. J. Org. Chem.* **27**, e202300891 (2024).